

WATER HARVESTING STRUCTURE AND PRACTICE

VAC 4 Science and Society Presentation

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INTRODUCTION

Water is one of the most important natural resources, yet its availability is decreasing day by day. Due to irregular rainfall, rapid population growth, urbanization, and excessive extraction of groundwater, many regions face serious water scarcity. A large amount of rainwater is wasted as surface runoff instead of being stored or reused. Water harvesting is a sustainable method that helps in collecting and conserving rainwater, reducing water shortage and supporting long-term water management.



WATER HARVESTING

Water harvesting refers to the process of collecting and storing rainwater for future use. It involves capturing rainwater from rooftops, open spaces, or catchment areas instead of allowing it to flow away as runoff. The stored water can be used for domestic needs, agriculture, and groundwater recharge, helping in the efficient management of water resources.

The main objective of water harvesting is to conserve rainwater and use it efficiently. It helps in recharging groundwater levels and reducing the pressure on existing water sources. Water harvesting also minimizes soil erosion and flooding caused by excessive runoff. Overall, it aims to ensure water availability during dry seasons and promote sustainable water management.



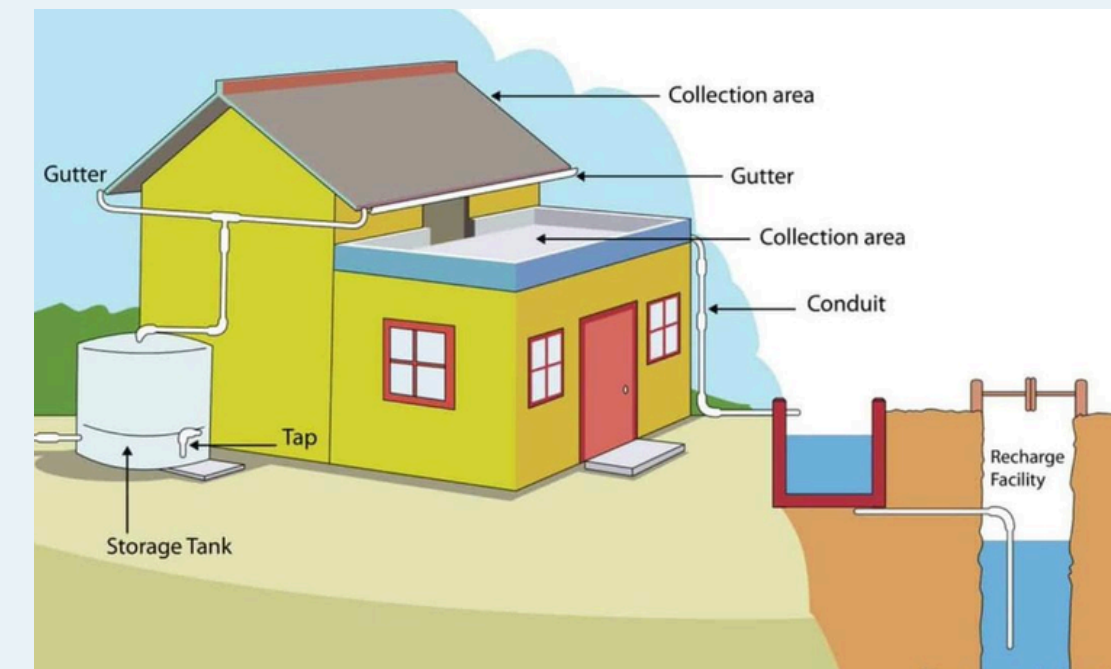
TYPES OF WATER HARVESTING

Rooftop Rainwater Harvesting

- Rooftop rainwater harvesting is a system used to collect rainwater falling on the roofs of buildings.
- Rainwater is first captured from the rooftop surface and directed through pipes or gutters.
- The collected water passes through filters to remove dust, leaves, and other impurities.
- Filtered water is stored in tanks or underground reservoirs for later use.
- Excess water can be diverted to recharge pits or recharge wells to replenish groundwater.
- This method is commonly used in urban and semi-urban areas where open land is limited.
- Rooftop water harvesting helps meet domestic water needs such as washing, cleaning, and gardening.
- It reduces dependence on municipal water supply and groundwater extraction.
- The system is cost-effective, easy to install, and requires minimal maintenance.
- Regular cleaning of roofs, pipes, and filters is essential for safe and efficient functioning.

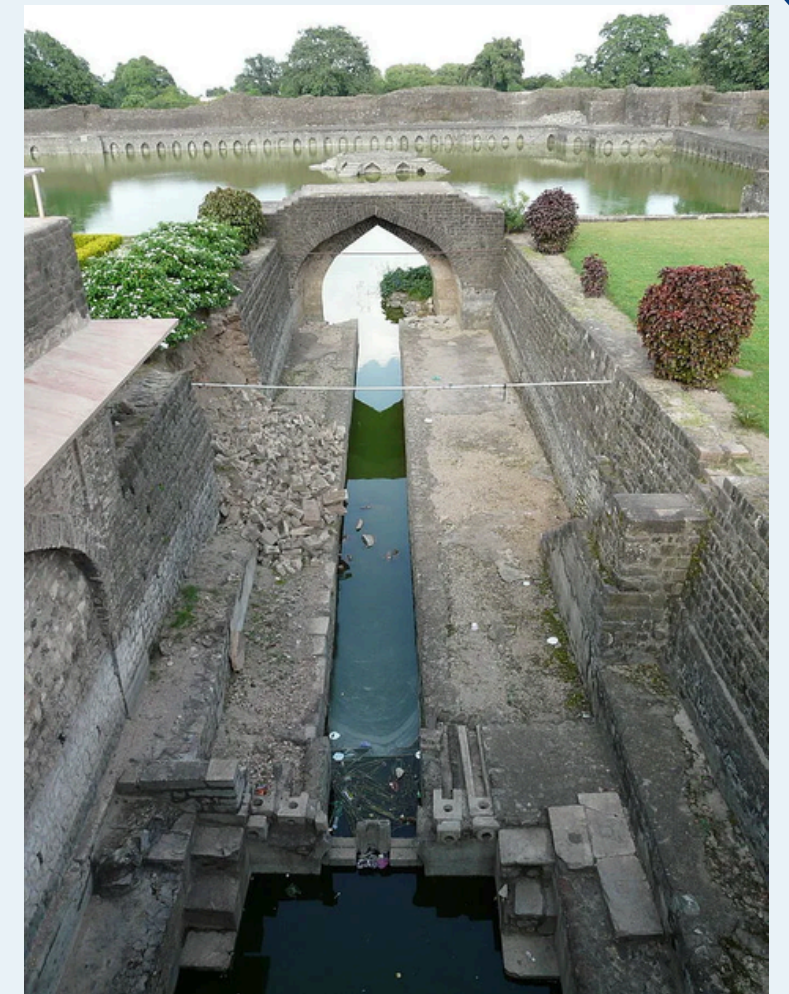
Surface Runoff Harvesting

- Surface runoff water harvesting involves collecting rainwater that flows over the land during rainfall.
- Rainwater from open fields, roads, and sloping land is directed towards storage or recharge structures.
- The collected water is stored in ponds, tanks, reservoirs, or small dams.
- This method is mainly practiced in rural and semi-arid areas.
- Stored water is used for irrigation, livestock, and sometimes domestic purposes after treatment.
- Surface runoff harvesting helps reduce soil erosion and flooding.
- It increases soil moisture and improves agricultural productivity.
- Structures like check dams, percolation tanks, and contour bunds are commonly used.
- Proper maintenance, desilting, and protection of catchment areas are essential for effectiveness.



Traditional Water Harvesting Structures

- Traditional water harvesting structures are ancient systems developed through indigenous knowledge
- These methods evolved based on local climate, rainfall patterns, soil type, and topography
- Constructed using locally available materials such as mud, stone, lime, and bricks
- Designed to collect, store, and conserve rainwater for long periods
- Help in reducing surface runoff and preventing soil erosion
- Play a major role in recharging groundwater and improving water tables
- Managed and maintained by local communities through collective efforts
- Support agriculture, domestic water needs, and livestock
- Environment-friendly, cost-effective, and sustainable in nature



A rainwater storage tank,
Mandu fort, Madhya Pradesh

Traditional Water Harvesting Structures

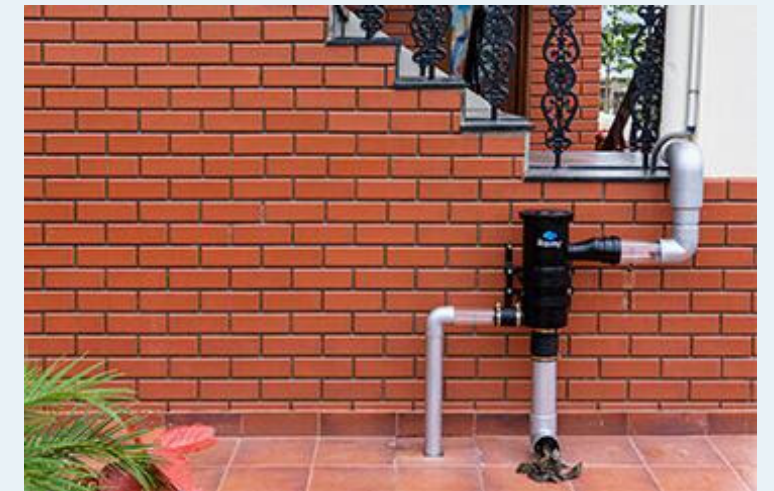
- Johads are small earthen embankments built to collect rainwater and recharge groundwater
- Baolis or stepwells are underground structures that store water and provide access during dry seasons
- Tanks and ponds are used to collect surface runoff for domestic use and irrigation
- Khadins are long earthen embankments used mainly in dry regions for agricultural water storage
- The ahar-pyne system consists of reservoirs and channels for irrigation management
- These traditional structures help reduce water scarcity and support sustainable agriculture



Agrasen Ki Baoli,
Delhi

Modern Water Harvesting Structures

- Modern water harvesting structures are designed using scientific, technical, and engineering principles
- These systems aim to collect rainwater efficiently and prevent it from flowing away as runoff
- They help control urban flooding and reduce waterlogging during heavy rainfall
- A major objective is to recharge groundwater and improve declining water table levels
- These structures are especially useful in urban and semi-urban areas with limited open space
- Materials such as concrete, PVC pipes, filters, and gravel are commonly used
- Proper planning, site selection, and technical design are required for effective functioning
- Regular inspection, cleaning, and maintenance are essential to ensure long-term success
- Modern structures support sustainable water use and long-term water security



PRACTICES OF WATER HARVESTING

- The first step is proper selection and protection of the catchment area
- Rooftops, open land, and channels should be kept clean for good quality water
- Rainwater should be guided properly through pipes or channels to storage areas
- Regular cleaning and removal of silt from ponds and tanks is necessary
- Small structures like bunds and check dams help slow down flowing water
- Rainwater can be directed to recharge pits and wells to refill groundwater
- Storage tanks and structures should be checked and repaired regularly
- People's participation helps in proper use and maintenance
- These practices help ensure water availability for a longer time

PYQ

Q1. The scarcity of clean water and decline in groundwater level is a significant concern. By briefly describing "Water Harvesting", explain its role in solving the problem of water scarcity.

PYQ

Ans. Water harvesting means collecting and storing rainwater or surface water for future use, instead of letting it run off. It can be done using methods like rooftop tanks, ponds, check dams, or percolation pits, which also help water soak into the ground.

Role in Solving Water Scarcity:

1. Increases groundwater levels by letting water percolate into the soil.
2. Provides extra water for drinking, irrigation, and daily use.
3. Reduces water runoff and prevents soil erosion.
4. Ensures water availability during dry periods.
5. Supports sustainable and environment-friendly water management.

By collecting and storing water, water harvesting helps reduce water scarcity and restores declining groundwater levels.

PYQ

Q2. Contrast ancient methods of water conservation and collection with contemporary water harvesting techniques. In what ways have both approaches supported sustainable management of water resources?

PYQ

Ans. Ancient methods of water conservation relied on structures like stepwells, tanks, reservoirs, and check dams to store water and recharge groundwater. Underground channels such as qanats transported water over long distances with minimal loss, and rainwater was often collected from roofs or small ponds for drinking, irrigation, and livestock. These approaches were low-tech, used local materials, and were managed by communities, blending with the natural environment. Modern water harvesting techniques, on the other hand, include rooftop rainwater harvesting with filtration, recharge wells, percolation tanks, stormwater collection in cities, and even desalination and treatment plants. They use engineering, technology, and scientific planning to meet the demands of urban and industrial areas. Both ancient and modern methods support sustainable water management by conserving water, recharging groundwater, and reducing wastage. While ancient techniques focused on small-scale, community-driven solutions, modern methods scale up these principles to serve larger populations efficiently, showing that both approaches play important roles in managing water resources responsibly.

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THANK YOU